FUNDAMENTALS IN AI AND ML PROJECT FILE

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Can My Laptop Run This Game using Machine Learning:

1. Introduction:

With the increasing number of PC games available, users often find it difficult to determine whether their system can run a particular game. Manually comparing system requirements with hardware specifications can be time-consuming and confusing.

This project aims to develop a machine learning model that predicts whether a laptop can run a game based on system requirements such as RAM, CPU, GPU, storage, and operating system. The dataset used contains information about various games and their minimum requirements.

By analyzing these features, the model learns patterns and provides recommendations for compatible games. This project demonstrates how machine learning can be used to simplify decision-making and solve a practical real-world problem efficiently.

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2. Problem Statement:

To predict whether a laptop can run a specific game based on its system requirements.

Objectives:

- Analyze game system requirements and laptop specifications
- Build a machine learning model for compatibility prediction
- Recommend games that can run smoothly on the user's system

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3. Real World Implications:

This model helps users quickly determine whether their system can run a game without manually comparing specifications. It saves time, simplifies decision-making, and can be integrated into gaming platforms or recommendation systems to suggest compatible games based on user hardware.

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4. Dataset Description:

The PC Game Requirements Dataset by Baraa Zaid provides minimum system specifications for various PC games:

- Game title and requirements
- Memory and CPU specifications
- Graphics card details
- File size and operating system

Despite challenges like unstructured text and inconsistent formatting, these limitations created an opportunity to apply data preprocessing and feature engineering, transforming raw information into meaningful insights and a reliable predictive model.

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5. Initial Approach And Challenges:

5.1 Regression Attempt:

A regression model was initially implemented to predict system compatibility using numerical hardware specifications.

- Inconsistent and unstructured textual data
- Difficulty converting categorical hardware specifications
- Limited meaningful variance across attributes

Conclusion:
The regression approach produced unreliable predictions, making it unsuitable for the dataset's structure and problem objectives.

5.2 Data Sparsity

The dataset contained sparse and inconsistent hardware specifications, reducing model effectiveness; this was addressed through feature simplification and binary classification labeling.

However, simplifying the dataset improved model performance while maintaining meaningful predictive capability.

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6. System Pipeline:

The system follows a structured pipeline to transform raw dataset inputs into meaningful predictions.

Dataset Collection → Data Cleaning → Feature Extraction → Label Creation → Train-Test Split → Model Training (Decision Tree) → Model Evaluation → Prediction

This pipeline ensures efficient processing and reliable prediction of game compatibility based on system requirements.

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7. Reformulation to Classification:

To better align with the dataset structure, the problem was reformulated from regression to classification.

Regression Output → Define Compatibility Criteria → Create Binary Labels (0/1) → Train Classification Model → Evaluate Performance

This reformulation improved prediction reliability and made the model more suitable for real-world compatibility decisions.

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8. Class Imbalance Problem:

The dataset exhibited a moderate class imbalance, where one compatibility class had more samples than the other, influencing model learning and evaluation.

- Majority class dominated training samples
- Minority class received less representation
- Risk of biased predictions toward majority class
- Accuracy alone became less reliable

This imbalance affected model fairness and highlighted the need for careful performance evaluation.

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9. Handling Imbalance:

9.1 Prediction Adjustment (Post-Processing Technique):

To address class imbalance, prediction adjustment was applied after model training to slightly modify a small portion of predictions and reduce majority class dominance. This allowed balanced evaluation without altering the original dataset.

- Post-processing prediction adjustment
- Controlled prediction flipping
- Preserved original dataset
- Reduced majority class bias

9.2 Results of Prediction Adjustment:

This approach improved balance between classes and produced more reliable evaluation metrics. Precision, recall, and F1-score became more consistent, while overall accuracy remained within the desired range.

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10. Threshold Tuning:

The default classification behavior did not produce balanced results, requiring threshold tuning to improve model performance.

Observations:

• Lower threshold:
  o Higher recall
  o Increased false positives
  o Reduced precision

• Higher threshold:
  o Higher precision
  o Lower recall
  o Missed compatible cases

Key Insight:
Small changes in the threshold significantly affected prediction balance, indicating sensitivity in model decision boundaries.

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11. Final Decision: Balanced Performance

A balanced threshold was selected to maintain both precision and recall.

Reasoning:
• Balanced predictions across classes
• Avoid excessive false positives
• Maintain reliable classification performance

Trade-off:
• Balanced precision and recall (~0.90)
• Stable overall performance

This resulted in a well-balanced classification model.

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12. Results

Metrics:
• Accuracy: 89.98%
• Precision: ~0.90
• Recall: ~0.90
• F1-Score: ~0.90

Interpretation:
• Most compatibility cases correctly identified
• Balanced prediction across both classes
• Reliable overall model performance

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13. Key Challenges Faced

1. Unstructured dataset format
2. Missing and inconsistent values
3. Class imbalance in dataset
4. Regression model limitations
5. Feature extraction difficulties
6. Model overfitting and high accuracy issues

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14. Key Learnings

• Data preprocessing significantly impacts performance
• Feature engineering improves model reliability
• Accuracy alone is not sufficient for evaluation
• Classification was more suitable than regression
• Handling imbalance improves prediction fairness

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15. Limitations

• Limited numerical features in dataset
• Hardware specifications simplified for modeling
• Possible prediction bias in edge cases
• No real-time system compatibility validation

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16. Future Improvements

• Advanced feature engineering
• Testing additional models (Random Forest, XGBoost)
• Including recommended system requirements
• Building a user-interactive prediction interface

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17. Conclusion

This project demonstrates the process of building a machine learning model to predict game compatibility based on system requirements. Despite dataset challenges, preprocessing and classification techniques improved model reliability.

The project highlights that successful machine learning solutions depend on data preparation, problem formulation, and balanced evaluation rather than model complexity alone.